

23MCT406

ENVIRONMENTAL SCIENCE

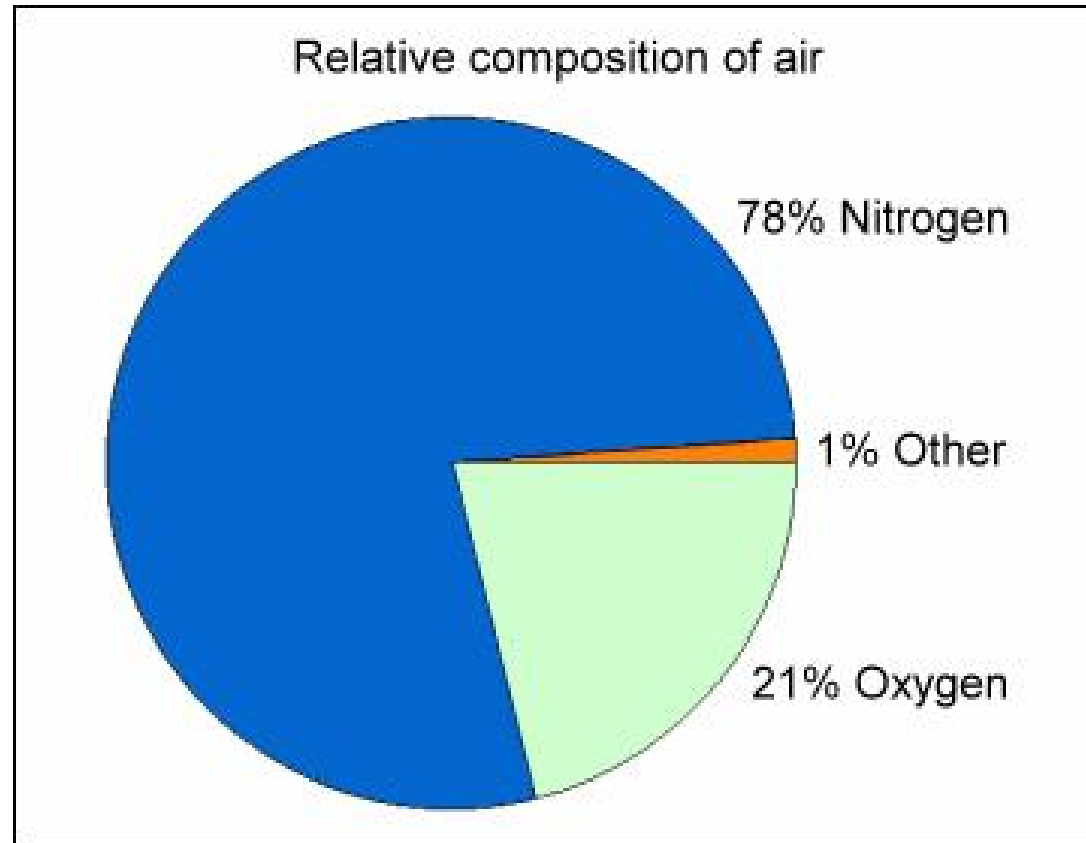


MODULE 2 ~ Air and Noise pollution

- Air pollutants – classification, sources and impacts
- Clean air act and national ambient air quality standards (NAAQS)
- Air quality index
- Emission reduction strategies ~ Understanding and controlling indoor air pollution.
- Ground level ozone and photochemical smog ~ Ozone layer depletion and the Montreal Protocol
- Global warming
- Noise Pollution: Sources and effects of noise
- Quantification of noise pollution (Leq, LAeq, etc.); Control and regulation rules in India

AIR

- **Air** is one of the most important constituent of our environment.
- Composition of air consist of:



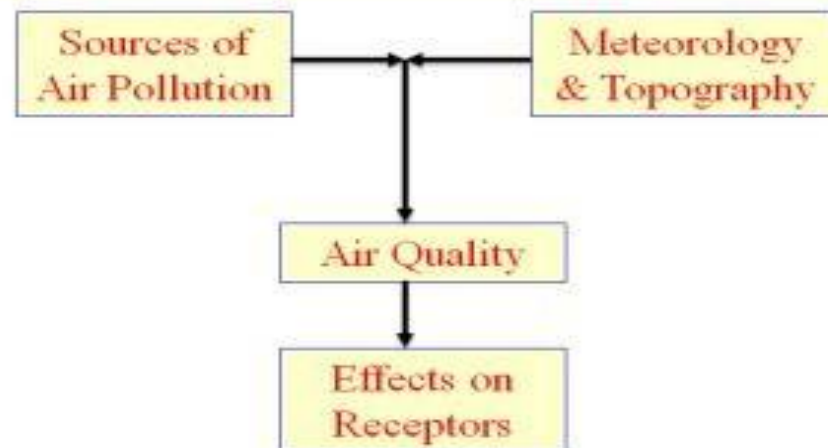
AIR POLLUTION

- Air pollution is defined as 'the presence of any air pollutant in the atmosphere.'
- They tend to **be injurious** to human beings or other living creatures or plants or property
- Air Pollution occurs when one or several air pollutants are present in such amounts for such a long period in the outside air



Air pollution definition

- WHO definition *“Air pollution is contamination of the indoor or outdoor environment by any chemical, physical or biological agent that modifies the natural characteristics of the atmosphere. Household combustion devices, motor vehicles, industrial facilities and forest fires are common sources of air pollution.”*
- EPA definition *“Air pollution is one or more chemicals or substances in high enough concentrations in the air to harm humans, other animals, vegetation, or materials. Such chemicals or physical conditions (such as excess heat or noise) are called air pollutants.”*



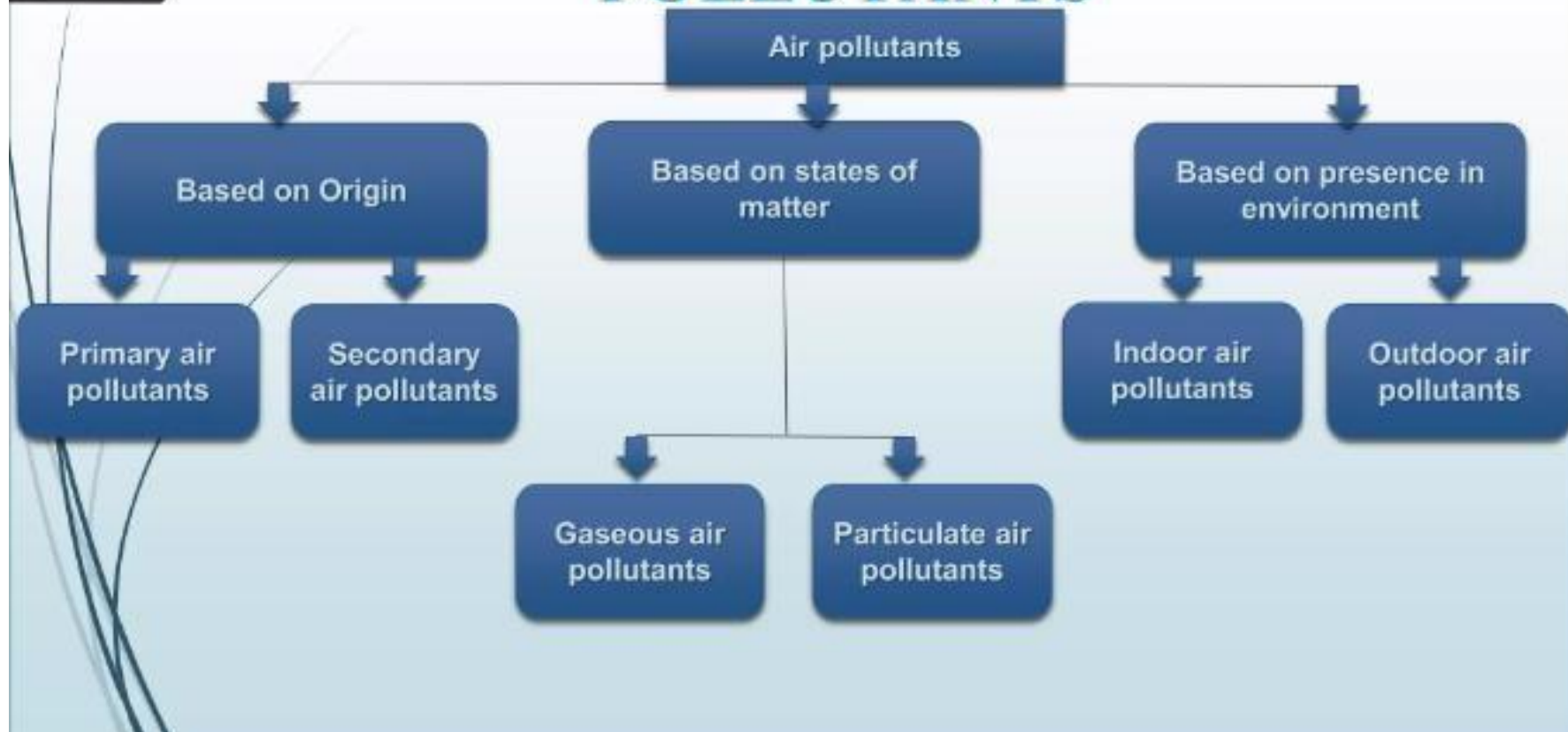
- Indian Standards (IS-4167 (1996)) defines air pollution as:

“Air pollution is the presence of foreign matter in the ambient atmosphere, generally resulting from the activity of man, in sufficient concentration, present for sufficient time and under circumstances which interfere significantly with the comfort, health or welfare of person or with the full use or enjoyment of property”

AIR POLLUTANTS

Air pollutant' has been defined as 'any solid, liquid or gaseous substance present in the atmosphere in such concentration

CLASSIFICATION OF AIR POLLUTANTS



Classification of Air Pollutants

a) Based on Origin

1. Primary Pollutants

- The pollutants that **directly cause air** pollution are known as primary pollutants
- Important primary pollutants are:
 - Carbon compounds, such as CO, CO₂, CH₄, and VOCs
 - Nitrogen compounds, such as NO, NO₂, and NH₃
 - Sulphur compounds, such as H₂S and SO₂
 - Halogen compounds, such as chlorides, fluorides, and bromides
 - Particulate Matter (PM or “aerosols”), either in solid or liquid form, categorized based on the aerodynamic diameter, such as PM10, PM2.5 etc.

Primary Pollutants

1. Carbon Monoxide (CO)

- Colorless, odorless and poisonous gas
- It is produced by the **incomplete burning of carbon-based fuels** (coal, petrol, diesel and wood)
- It comes from the automobile industries, exhaust devices
- About 70% of CO emissions are from the **transport sector**
- When the air is polluted with CO, human blood is likely to be **deprived of oxygen and leads to coma and death**
- **CO** has a strong affinity towards hemoglobin of blood stream & hence inhaled can be toxic to human beings
- In mild dosages, it leads to headache

2. Oxides of Sulphur (SO_x)

Sulphur dioxide (SO₂)

- SO₂ is a colorless gas
- It can be easily dissolved in water
- **SO₂** is the most common air pollutant
- SO₂ is a gas produced from **burning of coal, mainly in thermal power plants**
- Some industries such as paper mills produce SO₂
- SO₂ along with other impurities in the air can often be seen as **gray industrial smog** near industrial areas
- When **water molecules** in the atmosphere combine with **SO₂ and SO₃ acid rain is formed**
- It is injurious not only to men and plants, but it also attacks rapidly a few rocks such as limestone, marbles, electric contacts etc.
- This **corrodes** metals, damages buildings and monuments
- Also sulphate particles are capable of penetrating deep into the lungs

Contd...

Sulphur trioxide (SO_3)

- This is more irritant than SO_2 because it combines immediately with water to form sulphuric acid

Before and After effects of Acid Rain



Taj Mahal

3. Oxides of Nitrogen (NO_x)

Nitrogen dioxide (NO₂)

- Nitrogen dioxide (NO₂) is the most common air pollutant (which is **reddish brown in colour**)
- Major sources of NO_x, are **automobile exhaust and nitric acid (HNO₃) manufacturing industries**
- NO₂ along with other impurities in the air can often be seen as a reddish-brown layer called brown photo chemical smog) over many urban areas
- When water molecules in the atmosphere combines with NO₂, acid rain is formed
- This corrodes metals, damages buildings and monuments
- At higher concentration NO₂ may irritate the respiratory system **leading to cough and nasal congestions**

4. Chloro Fluoro Carbons (CFC's)

- CFC's (also known as **Freon**) are non- toxic
- They contain **Carbon, Fluorine and Chlorine atoms**
- The main CFCs are the following:
 - CFC – 11 (Trichloro Fluoro Methane CFCl_3)
 - CFC – 12 (Dichloro Fluoro Methane CF_2Cl_2)
- The major uses of CFCs are as coolants in refrigerators and in air conditioners, as solvents in cleaners particularly for electronic circuit boards etc.
- CFCs are the main cause of **ozone depletion**

2. Secondary Pollutants

- The pollutants formed by the intermingling and reaction of primary pollutants are known as secondary pollutants
- Examples are Sulphuric acid, Ozone, Formaldehyde, Peroxy-acetyl-nitrate (PAN), Photochemical smog etc.

1. Groundlevel Ozone (GL O₃)

- Groundlevel Ozone is not usually emitted directly into the air, but at the ground level it is created by a chemical reaction between NO_x and VOCs (Volatile Organic Compounds) in the presence of sunlight
- Ozone consists of oxygen molecules which contain three oxygen atoms
- It is not emitted directly into the air but produced in the atmosphere when oxygen combines with oxygen radical (O) in the presence of sunlight
- Groundlevel Ozone may irritate the respiratory system leading to nose and throat infections and can penetrate much deep into the lungs
- In plants, it causes **premature ageing and suppressed growth**

2) Smog

- Smog is a combination of **smoke and fog** or various gases when react in the presence of sunlight
- The effects of smog on human health cause for respiratory, irritation to the eyes, diseases related to nose, throat, bronchitis, pneumonia, headache, nerves, liver, and kidneys

Contd...

Two types of smog:

- a. Classical smog/London smog: It occurs in **cool humid climate**. It was first observed in 1952 in London. It is a mixture of **smoke, fog and sulphur dioxide**. Chemically it is a reducing mixture and so it is called as reducing smog
- b. Photochemical smog/Los Angeles smog: It occurs in **warm, dry and sunny climate**. It was first observed in 1940s in Los Angeles. The main components of the photochemical smog result from the **action of sunlight on unsaturated hydrocarbons & nitrogen oxides produced by automobiles and factories**. It has high concentration of oxidizing agents and is, therefore, called as oxidizing smog

3) Acid rain

- Acid rain has become one of the most important global environmental problems and poses significant adverse impact on soils, rivers, lakes, forests and monuments
- The phenomenon occurs when **SO_x and NO_x** from the **burning of fossil fuels** such as Petrol, Diesel, Coal etc combine with water vapour in atmosphere and **fall as rain or snow or fog**
- Natural sources like volcanoes, forest fires, etc also contribute SO_x and NO_x
- Increased urban and industrial activities cause air pollution resulting in the rise of concentration of SO_x and NO_x
- **Sulphur dioxide and NO₂ combines with water vapour** in the atmosphere **produce sulphuric acid and Nitric acid respectively and results acid rain**

b) Based on Chemical Composition

1. ORGANIC

- Organic compounds containing carbon and hydrogen
- They may include carboxylic acids, ethers, esters, amines, etc.
- **Natural organic** of biological origin like pollen, bacteria etc
- **Synthetic organic** pollutant like manufactured products

2. INORGANIC

- These include carbon monoxide, carbon dioxide, sulphur oxides, hydrogen chloride etc.

c) Based on state of matter

- Gaseous air pollutants: These pollutants exist in a gaseous state at normal temperature and pressure

Eg: carbon dioxide, nitrogen dioxide, sulphur oxides etc.

- Particulate air pollutants: These are not gaseous substances

Eg: suspended droplets, solid particles or mixtures of the two

Contd...

Particulate Pollutants	Gaseous Pollutants	
1. Lead 2. Fly Ash 3. Metallic Oxides 4. Nanoparticles	1. Carbon monoxide (CO) 2. Carbon dioxide (CO₂) 3. Chlorofluorocarbons (CFCs) 4. Ozone (O₃) 5. Nitrogen oxide (NO_x) 6. Sulphur dioxide (SO₂)	• Volatile organic compounds (VOCs) • Benzene • Ethylene • Biological pollutants • Asbestos • Radon

Sources of Air Pollutants

1. Natural source

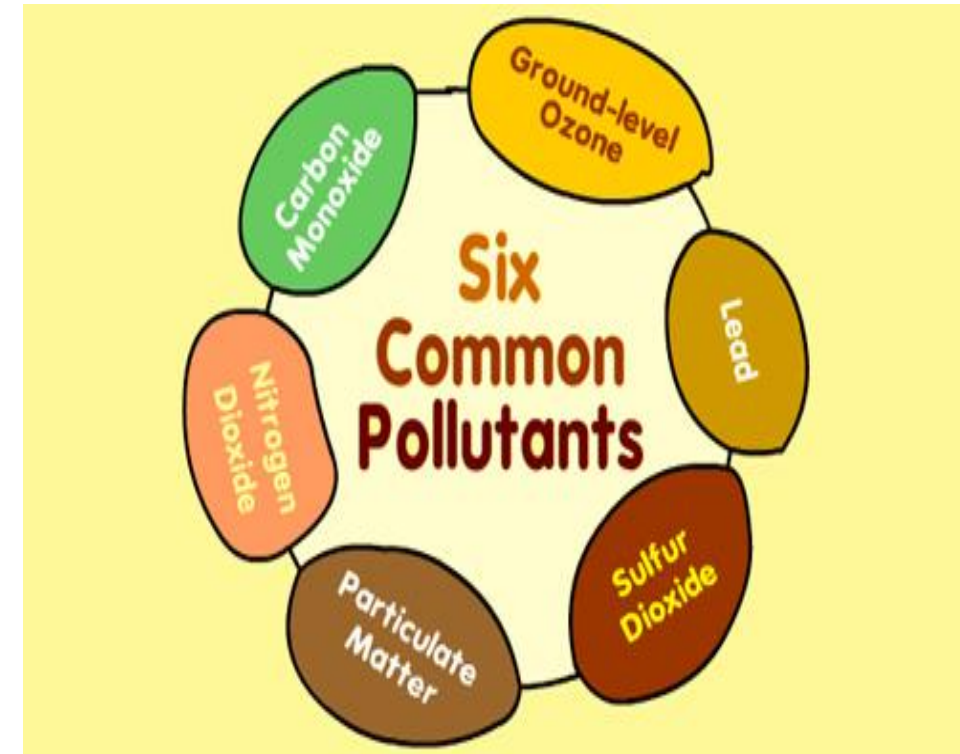
- **Volcanoes:** Volcanic activity produces smoke, ash, carbon dioxide, sulfur dioxide etc.
- **Dust:** Wind blown dust from areas with little or no vegetation such as desert areas.
- **Forest fires:** Forest fires created by natural causes, result in the formation and release of smoke, ash, dust, carbon dioxide, nitrogen oxides and other air pollutants.
- **Wetlands:** Microbial action in wetlands result in significant amounts of methane being formed and released to the atmosphere.

2. Anthropogenic sources

- **Stationary point sources:** A stationary point source is a single, identifiable source of air pollutant emissions. For example, the emissions from an industrial flue gas chimney / stack.
- **Mobile sources:** Mobile sources include the exhaust emissions from vehicles. A badly maintained car emits more pollution (up to 20% more volatile organic compounds and 10% more nitrogen oxides) compared to that of a well maintained one
- **Evaporative sources:** Evaporative sources are volatile liquids that, when not completely enclosed in a tank or other container, evaporate and release vapors over time. For example, liquids such as paints, solvents, pesticides, perfumes, hair sprays etc.

Outdoor & Indoor air pollutants

- **Outdoor air pollution** is primarily due to **automobile exhaust and industrial emissions**
- **Indoor air pollutants** are released by indoor activities such as **smoking, cooking, painting etc.**
- As per **United States Environmental Protection Agency (US EPA)**, the major outdoor air pollutants are known as **big six criteria air pollutants**
- The criteria pollutants are **carbon monoxide, lead, nitrogen dioxide, ozone, particulate matter, and sulphur dioxide**





CLASSIFICATION BASED ON PRESENCE IN ENVIRONMENT

- 1. Indoor air pollutants
- 2. Outdoor air pollutants

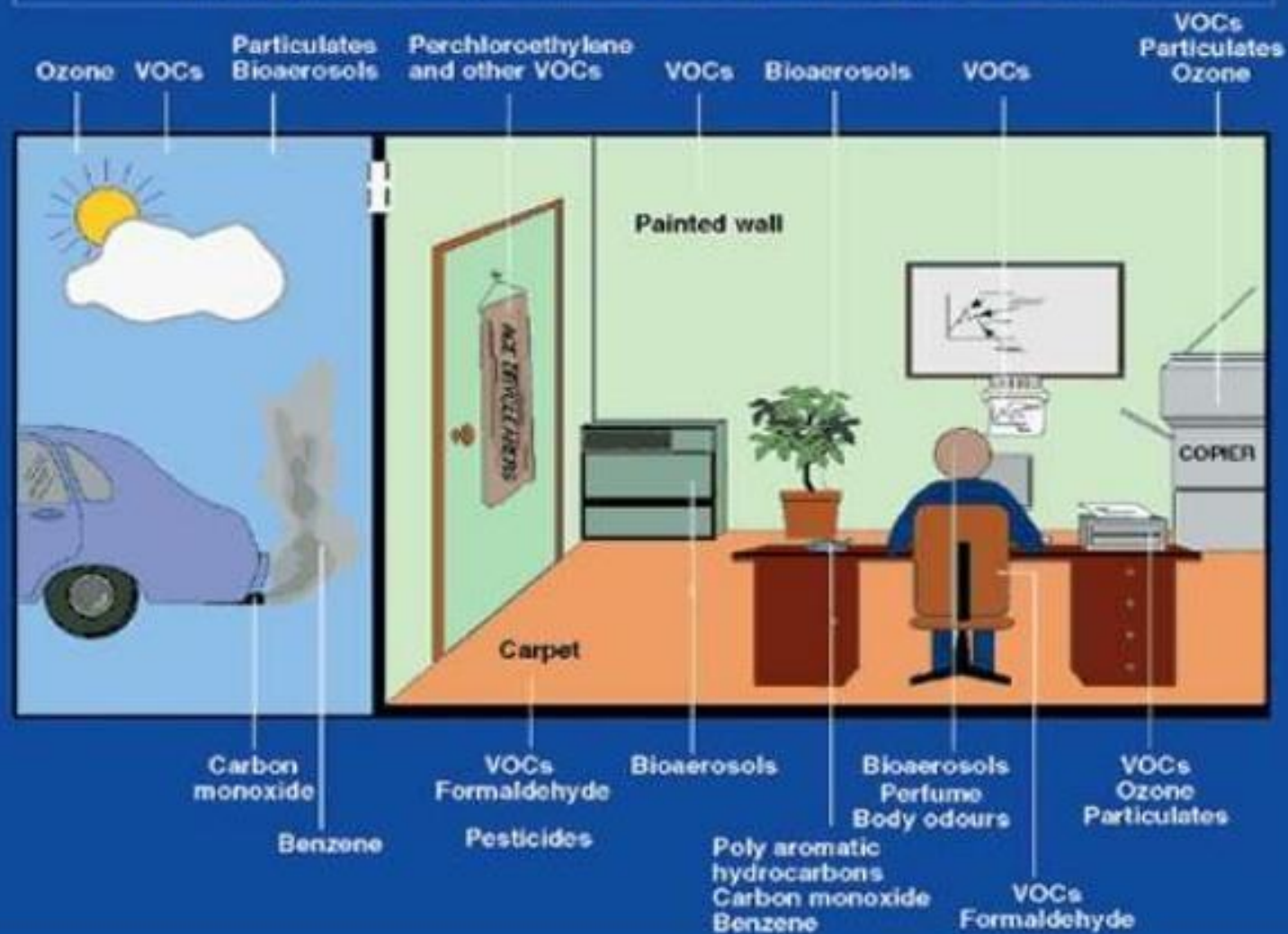


INDOOR AIR POLLUTANTS

- The air pollutants which are generated from households are called indoor air pollutants.
 - Cleaning agents
 - Mosquito repellents
 - Pesticides
 - Cigarette smoke
 - Gases from stoves etc.

Primary Sources of Indoor Air Pollution

- Outdoor air
- Building and Construction Materials and Furnishings
- Building Occupants and Activities
- Inadequate Building Design and Maintenance



INDOOR POLLUTION



➤ Outdoor air pollutants

- The air pollutants which are generated outside the buildings are called outdoor air pollutants.
 - Automobile pollutants
 - Industrial pollutants
 - Mining pollutants
 - Natural emissions from decaying matter and animals etc.

1. Carbon Monoxide (CO)

- Colorless, odorless and poisonous gas
- It is produced by the **incomplete burning of carbon-based fuels** (coal, petrol, diesel and wood)
- It comes from the automobile industries, exhaust devices
- About 70% of CO emissions are from the **transport sector**
- When the air is polluted with CO, human blood is likely to be **deprived of oxygen and leads to coma and death**
- **CO** has a strong affinity towards hemoglobin of blood stream & hence inhaled can be toxic to human beings
- In mild dosages, it leads to headache

2. Groundlevel Ozone (GL O₃)

- Groundlevel Ozone is not usually emitted directly into the air, but at the ground level it is created by a chemical reaction between NO_x and VOCs (Volatile Organic Compounds) in the presence of sunlight
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- Groundlevel Ozone may irritate the respiratory system leading to nose and throat infections and can penetrate much deep into the lungs
- In plants, it causes premature ageing and suppressed growth

3. Lead (Pb)

- Lead is a toxic material which is emitted into the atmosphere from leaded gasoline, leaded paint, lead smelters etc
- Lead is a cumulative poison and is more dangerous to children under the age of six
- Lead accumulation in the body affects the brain, causes mental retardation liver and kidney damage as well

4. Sulphur dioxide (SO₂)

- SO₂ is a **colorless gas**
- It can be easily dissolved in water
- **SO₂** is the most common air pollutant
- SO₂ is a gas produced from **burning of coal, mainly in thermal power plants**
- Some industries such as paper mills produce SO₂
- SO₂ along with other impurities in the air can often be seen as **gray industrial smog** near industrial areas
- When water molecules in the atmosphere combine with **SO₂ acid rain is formed**
- It is injurious not only to men and plants, but it also attacks rapidly a few rocks such as limestone, marbles, electric contacts etc.
- This **corrodes** metals, damages buildings and monuments
- Also sulphate particles are capable of penetrating deep into the lungs

5. Particulate Matter (PM)

- PM is general term used for a **mixture of solid particles and liquid droplets** found suspended in air
- They include **dust, smoke, flyash, fumes, mist, fog, aerosol** etc.
- Among these particles those which are **less than 10 μm in size remain suspended** in the atmosphere for a long period of time and are of a major concern as they can cause harmful health effects on humans
- Thus, the most harmful ones are **PM₁₀** (coarser particles) and **PM_{2.5}** (finer particles)

6. Nitrogen dioxide (NO₂)

- Nitrogen dioxide (NO₂) is the most common air pollutant (which is **reddish brown in colour**)
- Major sources of NO_x, are automobile exhaust and nitric acid (HNO₃) manufacturing industries
- NO₂ along with other impurities in the air can often be seen as a reddish-brown layer called brown photo chemical smog) over many urban areas
- When water molecules in the atmosphere combines with NO₂, acid rain is formed
- This corrodes metals, damages buildings and monuments.
- At higher concentration NO₂ may irritate the respiratory system leading to cough and nasal congestions

Impact of Air Pollutants

- Air pollution is a major environmental problem that affects not only living things but also our material and property
- In general, the effects of air pollution can be classified as follows:
 1. Effects on human health
 2. Effects on plants
 3. Effects on animals & birds
 4. Effects on materials and property
 5. Effects on environment

1. Effects on human health

- The effects of air pollution on human health generally occur as a result of contact between the pollutants and the body.
- The **health depends on the concentration of contaminant present, time of exposure to the contaminant and the vulnerability of the person**
- Some of the common health effects are:
 - ❖ Eye irritation
 - ❖ Nose and throat irritation
 - ❖ Increase in mortality rate
 - ❖ Chronic pulmonary diseases like bronchitis and asthma [generally known as Chronic Obstructive Pulmonary Disease (COPD)]
 - ❖ Carbon monoxide readily combines with haemoglobin in blood thus replacing oxygen from blood
 - ❖ Carcinogenic agents cause cancer

2. Effects on plants

Some of the common effects on plants due to air pollutants are given below.

- Suppressed growth and premature ageing in plants
- Causes leaf bleaching which results in Chlorosis (Photosynthesis is affected due to loss of Chlorophyll)
- Premature falling of leaves (known as abscission)
- Causes necrosis (dead spots on the leaf structure)



Dead spot

3. Effects on animals & birds

Some of the common effects on animals due to air pollutants are given below:

- Affects the mucous lining of the respiratory tract
- Causes bronchitis and asthma
- Lack of appetite in pet animals
- Acid deposition cause aquatic life damage
- Migration of seasonal birds are hampered due to severe air pollution

4. Effects on materials and property

Various economic losses occur because of material damage due to air pollutants. Some of the common effects on material and property due to air pollutants are given below.

- Acid deposition can corrode metals, eat away stone on statues and monuments
- Discolor buildings, cloth fabrics

5. Effects on environment

- Severe air pollution reduces the visibility due to smog formation (In the presence of light, smoke from vehicles and industries react with fog to form smog)
- Air pollutants can travel long distances (i.e. air pollutants are carried away to distant places by the blowing wind) which results in global (transboundary") pollution
- Some of the common examples are climate change, acid rain, global warming etc.

(Pollutants that enter the environment at one place having its effects hundreds of thousand of kilometres away from the source is known as transboundary pollution)

CLEAN AIR ACT (CAA)

- The **Clean Air Act (CAA)** is a comprehensive federal law in the United States that was enacted to regulate air pollution and protect public health and the environment from harmful emissions
- Initially passed in **1963** and significantly amended in **1970, 1977, and 1990**, the CAA provides the framework for controlling air quality and reducing pollution at a national level.

Key Objectives of the Clean Air Act

1. Improve Air Quality

- Set and enforce standards to reduce air pollution.
- Protect against harmful effects of air pollutants on human health and the environment

2. Regulate Emissions

- Address emissions from both stationary sources (e.g., factories, power plants) and mobile sources (e.g., vehicles, airplanes)

3. Protect Public Health

- Establish limits for pollutants that can harm health, particularly for vulnerable populations like children, the elderly, and those with preexisting health conditions

4. Preserve the Environment

- Reduce pollutants that cause environmental issues such as acid rain, smog, and damage to ecosystems

NATIONAL AMBIENT AIR QUALITY STANDARDS (NAAQS)

- National Ambient Air Quality Standards (NAAQS) are standards for air quality that are set by the Central Pollution Control Board (CPCB) that are applicable all over the country
- NAAQS stands for **National Ambient Air Quality Standards**, which are the maximum levels of certain pollutants that are allowed in the air
- The standards are set to protect public health, property, and vegetation

NATIONAL AMBIENT AIR QUALITY STANDARDS (NAAQS)

The objectives of air quality standards are:

- To indicate the levels of air quality necessary with an adequate margin of safety to protect the public health, vegetation and property
- To assist in establishing priorities for abatement and control of pollutant level
- To provide uniform yardstick for assessing air quality at national level
- To indicate the need and extent of monitoring programme

* Ministry of Environment and Forests, Government of India notification, 1994

** Particle size less than 10 μm

AMBIENT AIR QUALITY STANDARDS

Pollutant	Time Weighted Average	Concentration in Ambient Air		
		Industrial Area	Residential, Rural and other	Sensitive Area
Sulphur Dioxide (SO_2)	Annual 24 hours	80 $\mu\text{g}/\text{m}^3$ 120 $\mu\text{g}/\text{m}^3$	60 $\mu\text{g}/\text{m}^3$ 80 $\mu\text{g}/\text{m}^3$	15 $\mu\text{g}/\text{m}^3$ 30 $\mu\text{g}/\text{m}^3$
Oxides of Nitrogen (NO_x)	Annual 24 hours	80 $\mu\text{g}/\text{m}^3$ 120 $\mu\text{g}/\text{m}^3$	60 $\mu\text{g}/\text{m}^3$ 80 $\mu\text{g}/\text{m}^3$	15 $\mu\text{g}/\text{m}^3$ 30 $\mu\text{g}/\text{m}^3$
Suspended Particulate Matter (SPM)	Annual 24 hours	360 $\mu\text{g}/\text{m}^3$ 500 $\mu\text{g}/\text{m}^3$	140 $\mu\text{g}/\text{m}^3$ 200 $\mu\text{g}/\text{m}^3$	70 $\mu\text{g}/\text{m}^3$ 100 $\mu\text{g}/\text{m}^3$
Respirable ** Particulate Matter (RPM)	Annual 24 hours	120 $\mu\text{g}/\text{m}^3$ 150 $\mu\text{g}/\text{m}^3$	60 $\mu\text{g}/\text{m}^3$ 100 $\mu\text{g}/\text{m}^3$	50 $\mu\text{g}/\text{m}^3$ 75 $\mu\text{g}/\text{m}^3$
Lead (pb)	Annual 24 hours	1.0 $\mu\text{g}/\text{m}^3$ 1.5 $\mu\text{g}/\text{m}^3$	0.75 $\mu\text{g}/\text{m}^3$ 1.00 $\mu\text{g}/\text{m}^3$	0.50 $\mu\text{g}/\text{m}^3$ 0.75 $\mu\text{g}/\text{m}^3$
Carbon Monoxide (CO)	8 hours 1 hour	5.0 $\mu\text{g}/\text{m}^3$ 10.0 $\mu\text{g}/\text{m}^3$	2.0 $\mu\text{g}/\text{m}^3$ 4.0 $\mu\text{g}/\text{m}^3$	1.0 $\mu\text{g}/\text{m}^3$ 2.0 $\mu\text{g}/\text{m}^3$

AIR QUALITY INDEX

- The National Air Quality Index (AQI) was launched in New Delhi on September 17, 2014 under the Swachh Bharat Abhiyan
- The Central Pollution Control Board along with State Pollution Control Boards has been operating National Air Monitoring Program (NAMP) covering 240 cities of the country having more than 342 monitoring stations
- There are **six AQI categories**, namely Good, Satisfactory, Moderately polluted, Poor, Very Poor, and Severe
- The proposed AQI will consider **eight pollutants (PM10, PM2.5, NO2, SO2, CO, O3, Pb and NH3)** for which short-term (up to 24-hourly averaging period) National Ambient Air Quality Standards are prescribed

AIR QUALITY INDEX

- The **Air Quality Index (AQI)** is a standardized system used to measure and report air pollution levels in an easily understandable format.
- It provides a numerical scale that indicates the cleanliness or pollution level of the air, along with associated health advisories.
- The Air Quality Index (AQI) plays a crucial role in assessing and communicating the quality of air in a specific area.
- It serves as a vital tool for governments, environmental agencies, and individuals to understand pollution levels and their potential health impacts.
- By providing real-time data on air quality, the AQI enables timely actions to mitigate pollution exposure, helping to safeguard public health and promote environmental sustainability.

IMPORTANCE OF AIR QUALITY INDEX

- The AQI ranges from **0 to 500**, where lower values signify cleaner air, and higher values indicate more polluted air.
- It takes into account various pollutants, such as:
 - Particulate matter (PM_{2.5} and PM₁₀)
 - Ozone (O₃)
 - Nitrogen dioxide (NO₂)
 - Sulfur dioxide (SO₂)
 - Carbon monoxide (CO)

- Each of these pollutants affects health differently, making AQI a comprehensive measure for assessing air quality.
- **a) Protecting Public Health**
- AQI plays a crucial role in alerting people about the risks of air pollution, especially for vulnerable groups such as children, the elderly, and individuals with respiratory issues.
- **b) Policy Making**
- Governments and environmental agencies use AQI data to create policies and implement measures to reduce air pollution.
- **c) Awareness and Behavior Change**
- It helps individuals understand the impact of their actions on air quality and encourages sustainable practices such as reducing vehicle emissions

COMPONENTS OF AQI

- The AQI is determined by analyzing several key pollutants that impact air quality.
- These components include particulate matter (PM_{2.5} and PM₁₀), ground-level ozone (O₃), carbon monoxide (CO), sulfur dioxide (SO₂), and nitrogen dioxide (NO₂). Each of these pollutants is assigned a value, and the highest value among them determines the overall AQI for a region.
- Understanding these components helps identify the sources of pollution and guides effective measures to reduce their levels.

HOW IS AQI CALCULATED?

- AQI is calculated using the concentration levels of pollutants in the air.
- Each pollutant's concentration is converted into a sub-index using a pre-defined scale, and the highest sub-index becomes the overall AQI for that location.

Key Steps in AQI Calculation:

- **Measurement of Pollutants:** Monitoring stations record pollutant levels.
- **Conversion to AQI Values:** Concentration levels are mapped to AQI categories using specific breakpoints.
- **Reporting:** The highest pollutant-specific sub-index determines the AQI for that day.

- The method for calculating the Air Quality Index (AQI) is standardized and consistent across both the Indian CPCB (Central Pollution Control Board) and the US-EPA (Environmental Protection Agency).
- The AQI is determined separately for each pollutant parameter, and the highest sub-index among them is used to represent the overall AQI. A sub-index is a linear function derived from the pollutant's concentration.

Reference:**National Air Quality Index, CPCB**

The general equation for the sub-index (I_i) for a given pollutant concentration (C_p); as based on 'linear segmented principle' is calculated as:

$$[4] \quad I_i = \left[\left\{ (I_{HI} - I_{LO}) / (B_{HI} - B_{LO}) \right\} \star (C_p - B_{LO}) \right] + I_{LO}$$

where,

B_{HI} = Breakpoint concentration greater or equal to given concentration.

B_{LO} = Breakpoint concentration smaller or equal to given concentration.

I_{HI} = AQI value corresponding to B_{HI}

I_{LO} = AQI value corresponding to B_{LO}

C_p = Pollutant concentration

$$AQI = \text{Max } (I_p) \text{ (where; } p = 1, 2, \dots, n; \text{ denotes } n \text{ pollutants)}$$

Table 1: Indian Air Quality Index breakpoints for different Pollutants

Proposed Breakpoints for AQI Scale 0-500
(units: $\mu\text{g}/\text{m}^3$ unless mentioned otherwise)

AQI Category (Range)	PM ₁₀ 24-hr	PM _{2.5} 24-hr	NO ₂ 24-hr	O ₃ 8-hr	CO 8-hr (mg/ m ³)	SO ₂ 24-hr	NH ₃ 24-hr	Pb 24-hr
Good (0-50)	0-50	0-30	0-40	0-50	0-1.0	0-40	0-200	0-0.5
Satisfactory (51-100)	51-100	31-60	41-80	51-100	1.1-2.0	41-80	201-400	0.5 –1.0
Moderately polluted (101-200)	101-250	61-90	81-180	101-168	2.1- 10	81-380	401-800	1.1-2.0
Poor (201-300)	251-350	91-120	181-280	169-208	10-17	381-800	801-1200	2.1-3.0
Very poor (301-400)	351-430	121-250	281-400	209-748*	17-34	801-1600	1200-1800	3.1-3.5
Severe (401-500)	430 +	250+	400+	748+*	34+	1600+	1800+	3.5+

**One hourly monitoring (for mathematical calculation only)*

AQI category based on the AQI value:

Good (0-50)	Satisfactory (51-100)	Moderately polluted (101-200)	Poor (201-300)	Very poor (301-400)	Severe (> 401)
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AQI	Associated Health Impacts
Good (0–50)	Minimal impact
Satisfactory (51–100)	May cause minor breathing discomfort to sensitive people.
Moderately polluted (101–200)	May cause breathing discomfort to people with lung disease such as asthma, and discomfort to people with heart disease, children and older adults.
Poor (201–300)	May cause breathing discomfort to people on prolonged exposure, and discomfort to people with heart disease.
Very poor (301–400)	May cause respiratory illness to the people on prolonged exposure. Effect may be more pronounced in people with lung and heart diseases.
Severe (401–500)	May cause respiratory impact even on healthy people, and serious health impacts on people with lung/heart disease. The health impacts may be experienced even during light physical activity.

Emission reduction strategies

Some of the control/ remedial measures to reduce air pollution are:

- Determine the priority air pollutants~ based on health effect and the severity of the air quality problem
- Stacks, gets diluted and dispersed at a higher altitude itself. This results in reduced concentration of pollutants at the ground-level)
- Promotion of public transport, cycling & walking for local travel
- Stop excessive idling of vehicles
- Incorporate the control measures into a plan with implementation dates
- Improving infrastructure on public transport as a Greenhouse Gas (GHG) emission reduction measure (decreases traffic congestion and travel time which results in fuel saving)

Contd...

- Involve the public (by consulting) in air quality management activities as a part of the strategy development process (This early consultation reduces later challenges and streamlines the process of implementation)
- Afforestation programme (plants absorb CO_2 during photosynthesis and releases O_2)
- Periodic air quality monitoring
- Adoption of stringent pollution control measures
- Reduction in the use of conventional fuels
- Promotion of use of renewable energy sources.
- Use of Diesel Particulate Filter (DPF) in vehicular exhaust system will not allow emission of PM 2.5 into the environment.

Contd...

- The most effective method to control air pollution is to **prevent the formation of air pollutants or to reduce the emission of air pollutants at the source itself**
- For this, the following strategies can be adopted:
 1. **Catalytic convertors are to be provided to the automobile exhaust to control emissions**
 2. **Adoption of low sulphur fuels in vehicles**
 3. **Use of Electrostatic precipitators, fabric filters and scrubbers (to remove particulate matter from waste gas streams)**
 4. **Use of sorption technique (either adsorption of impurity onto a solid surface or absorption of the impurity in a liquid) to remove gaseous pollutants**